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Maruyama

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(54) **LOWER VEHICLE-BODY STRUCTURE**

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Primary Examiner — Dennis H Pedder

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(74) Attorney, Agent, or Firm — Rankin, Hill & Clark LLP

(30) **Foreign Application Priority Data**

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(57) **ABSTRACT**

(51) **Int. Cl.**
B62D 25/20 (2006.01)
B62D 21/15 (2006.01)

A lower vehicle-body structure includes: a floor cross member coupling a side sill disposed on an outer side in a vehicle width direction of a vehicle body to a floor tunnel disposed in a center in the vehicle width direction of the vehicle body, and disposed so as to be held between seat brackets in a vehicle front-rear direction; and a tunnel reinforcing member provided in the floor tunnel, wherein the tunnel reinforcing member includes a first tunnel reinforcing member disposed at a position to overlap in the vehicle front-rear direction with the floor cross member, and second tunnel reinforcing members disposed at positions to overlap in the vehicle width direction at least partly with the seat bracket, and the seat brackets are coupled to the floor cross member.

(52) **U.S. Cl.**
CPC **B62D 25/2009** (2013.01); **B62D 25/20** (2013.01); **B62D 25/2036** (2013.01); **B62D 21/157** (2013.01)

(58) **Field of Classification Search**
CPC B62D 25/20; B62D 25/025; B62D 25/04; B62D 25/2009; B62D 25/2036; B62D 21/157
USPC 296/193.07, 209, 204, 203.03
See application file for complete search history.

7 Claims, 13 Drawing Sheets

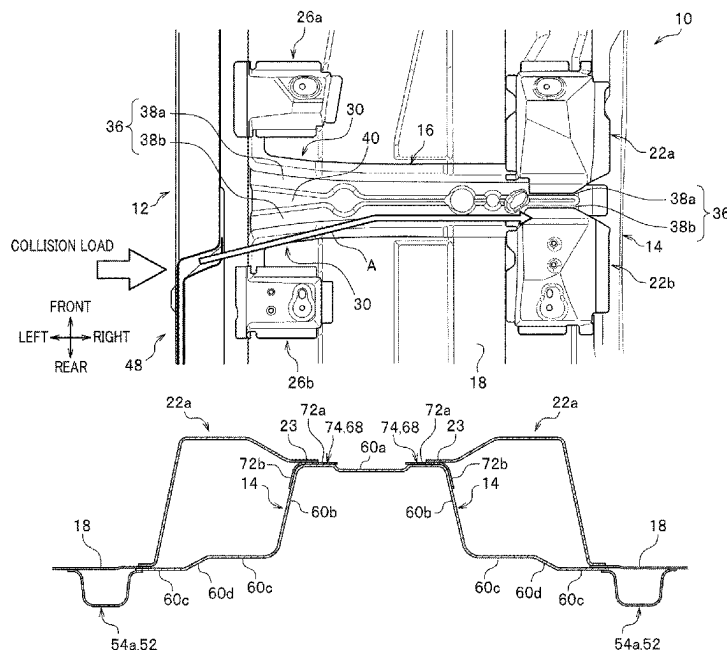


FIG. 1

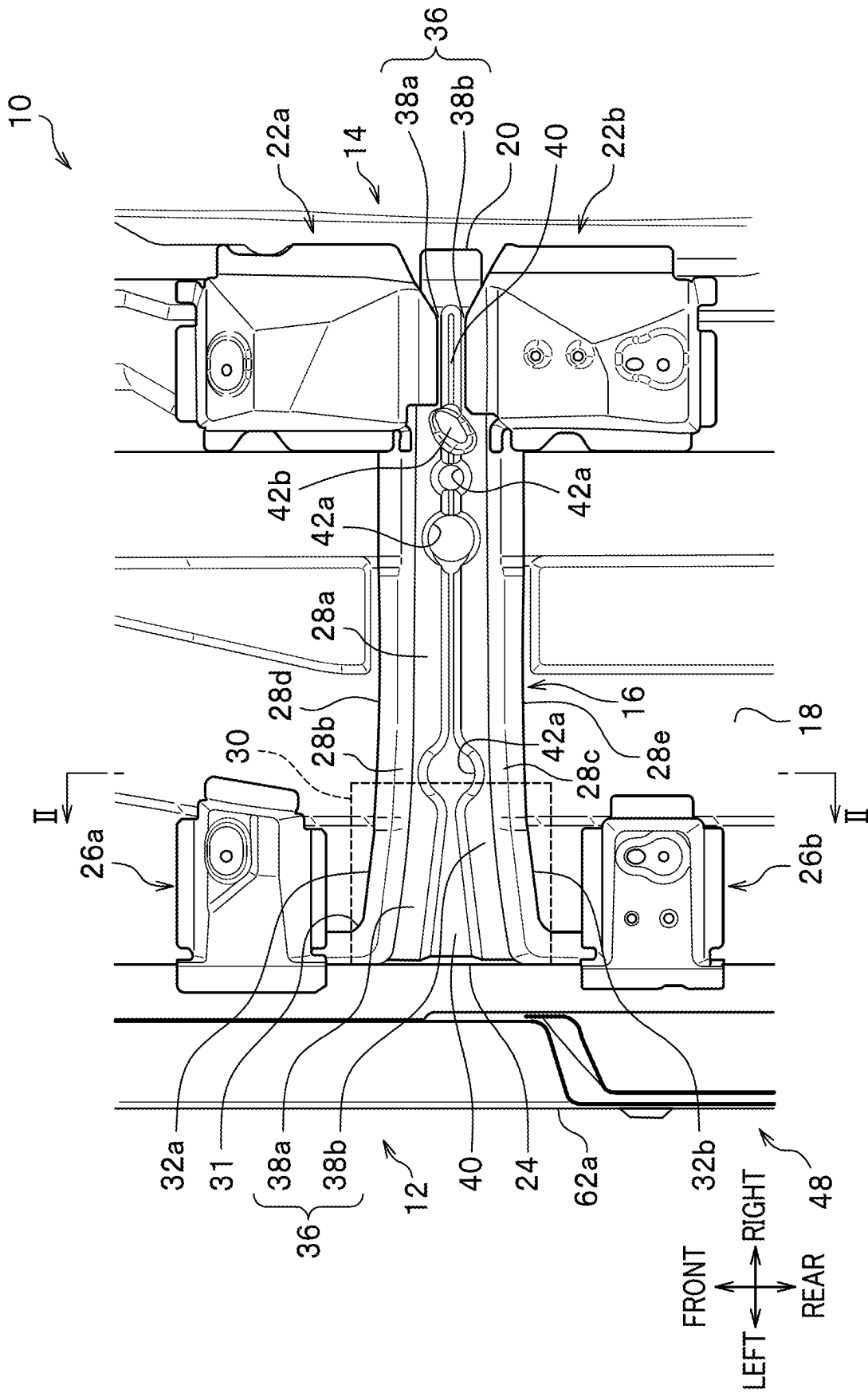


FIG. 3

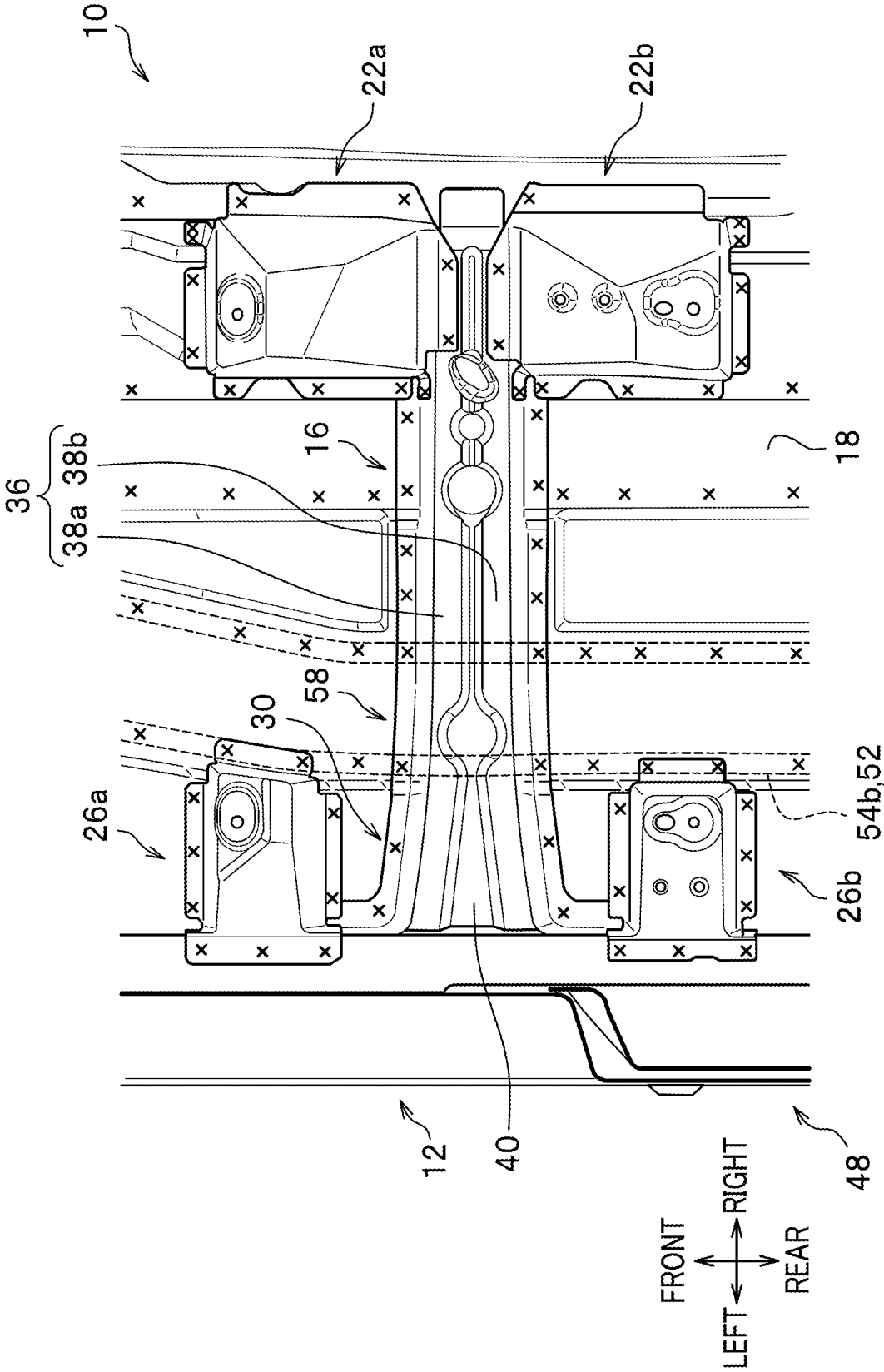


FIG. 4

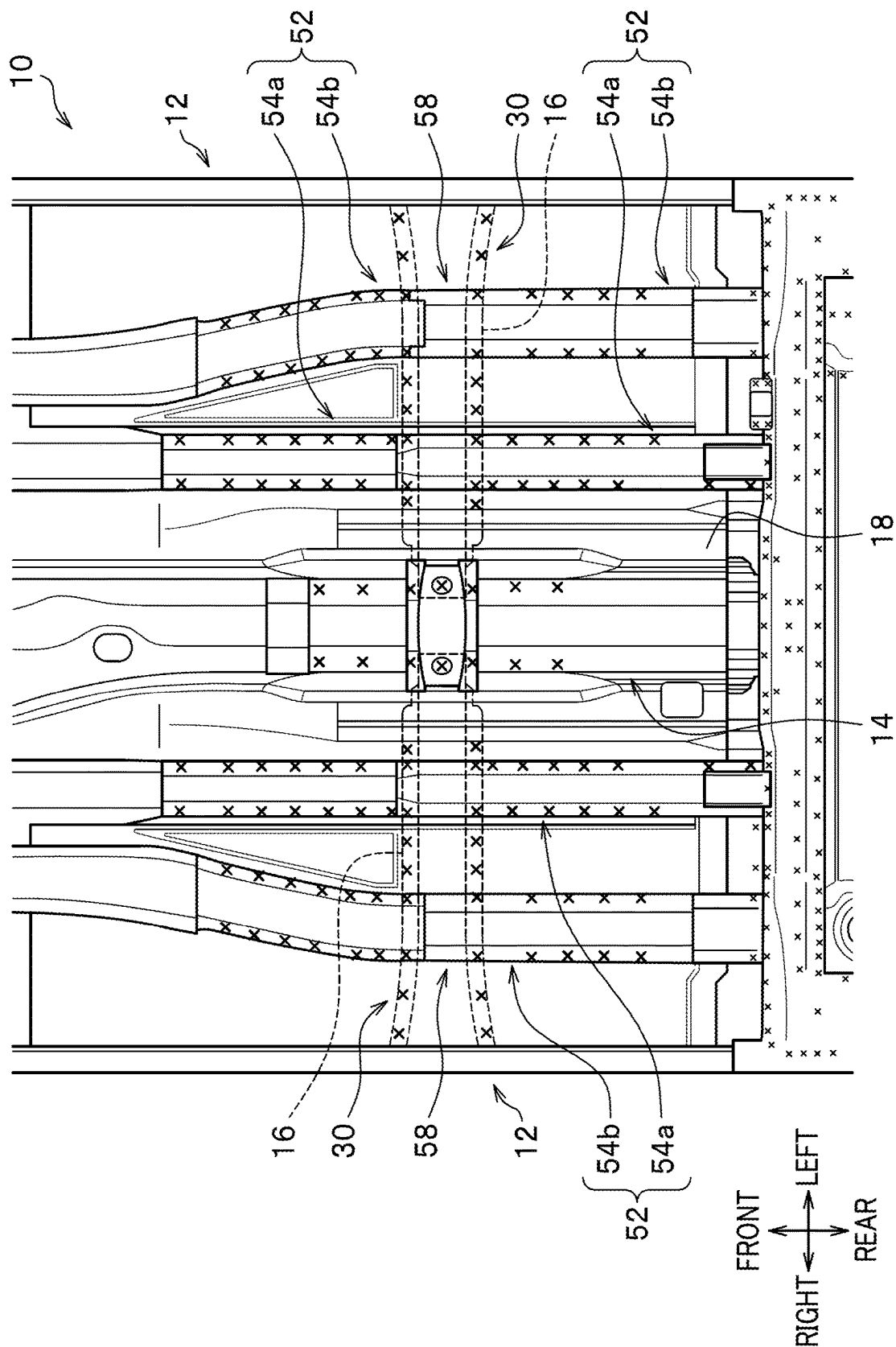


FIG. 5

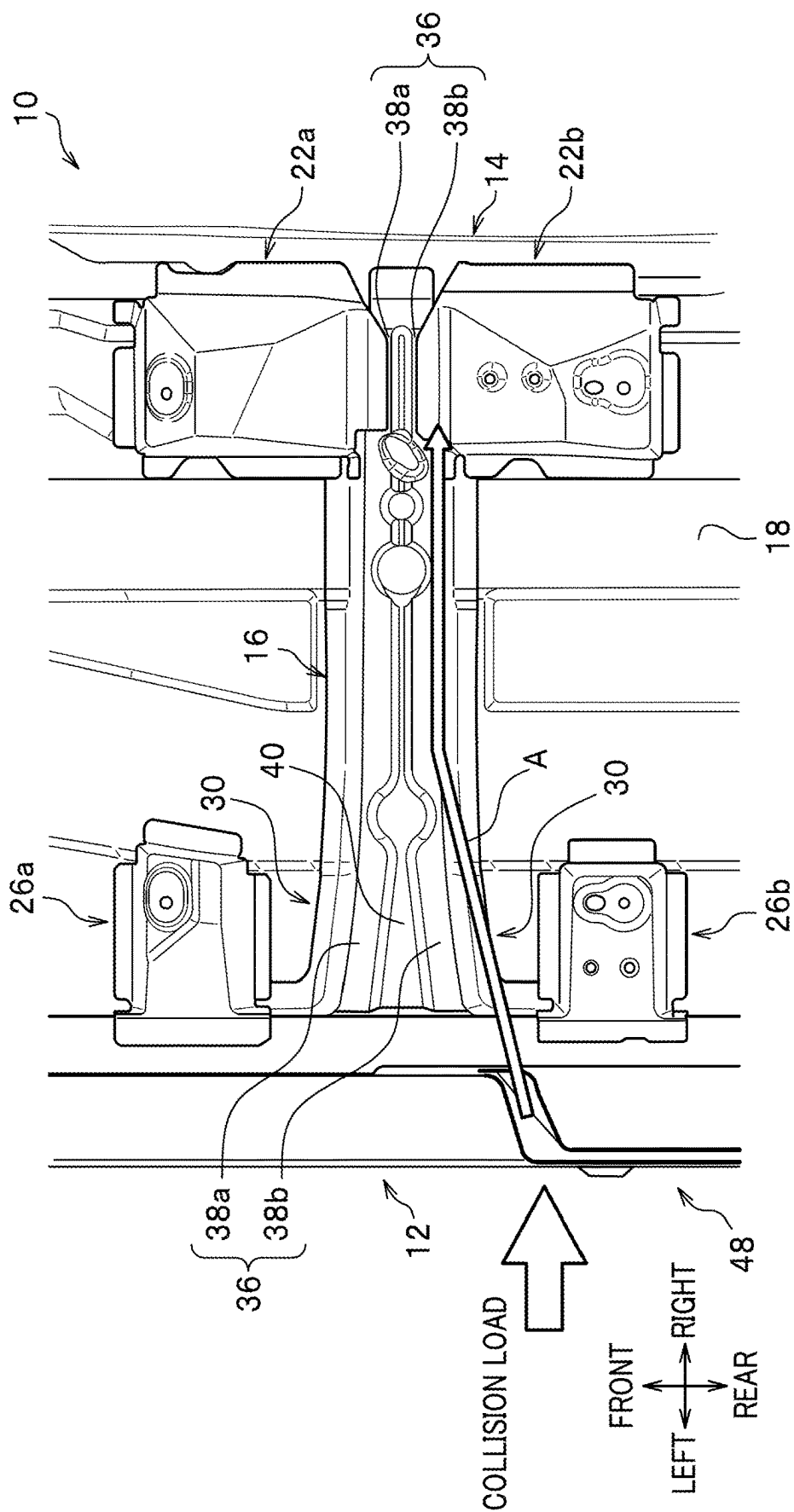


FIG. 6

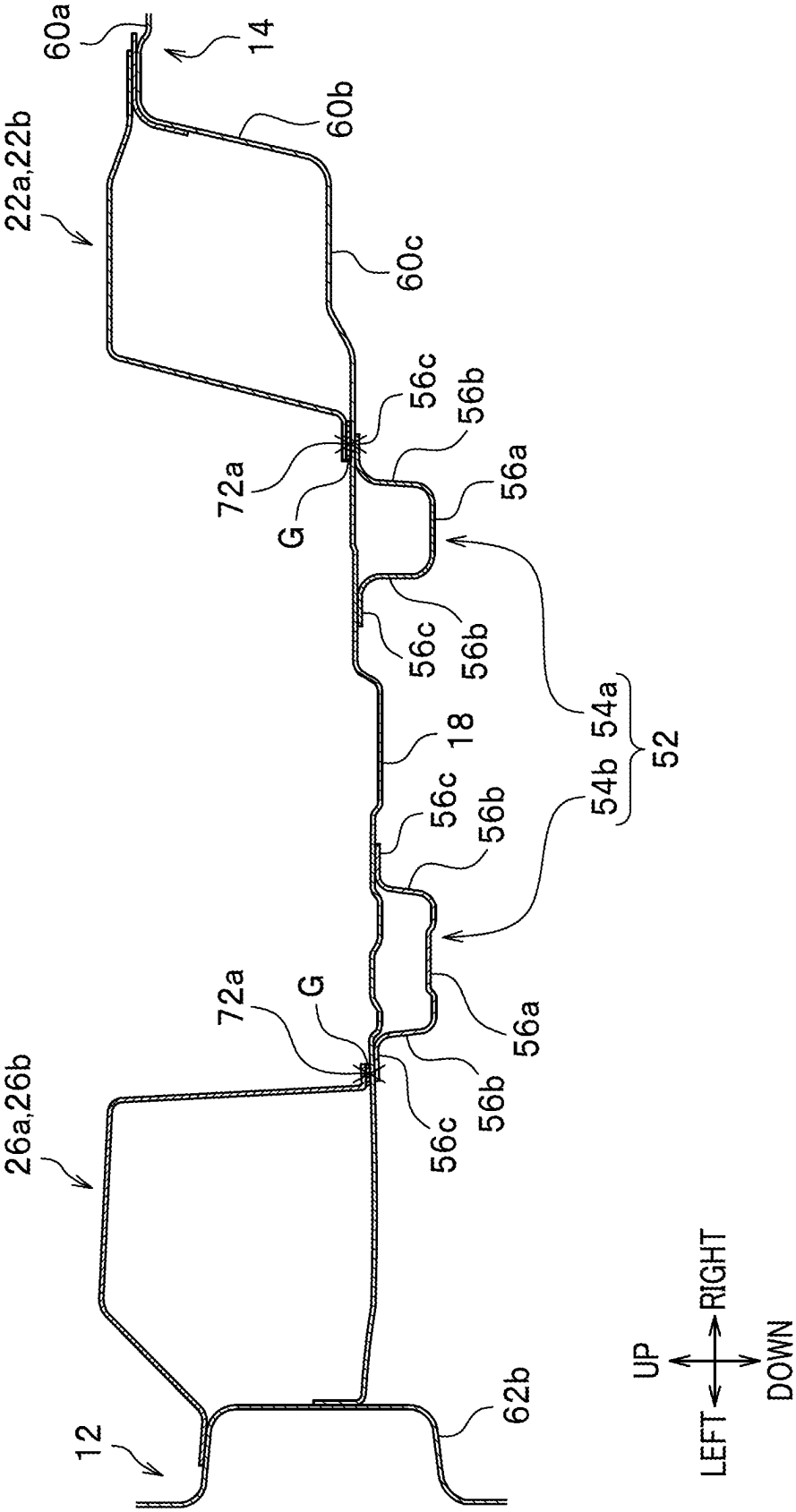


FIG. 7

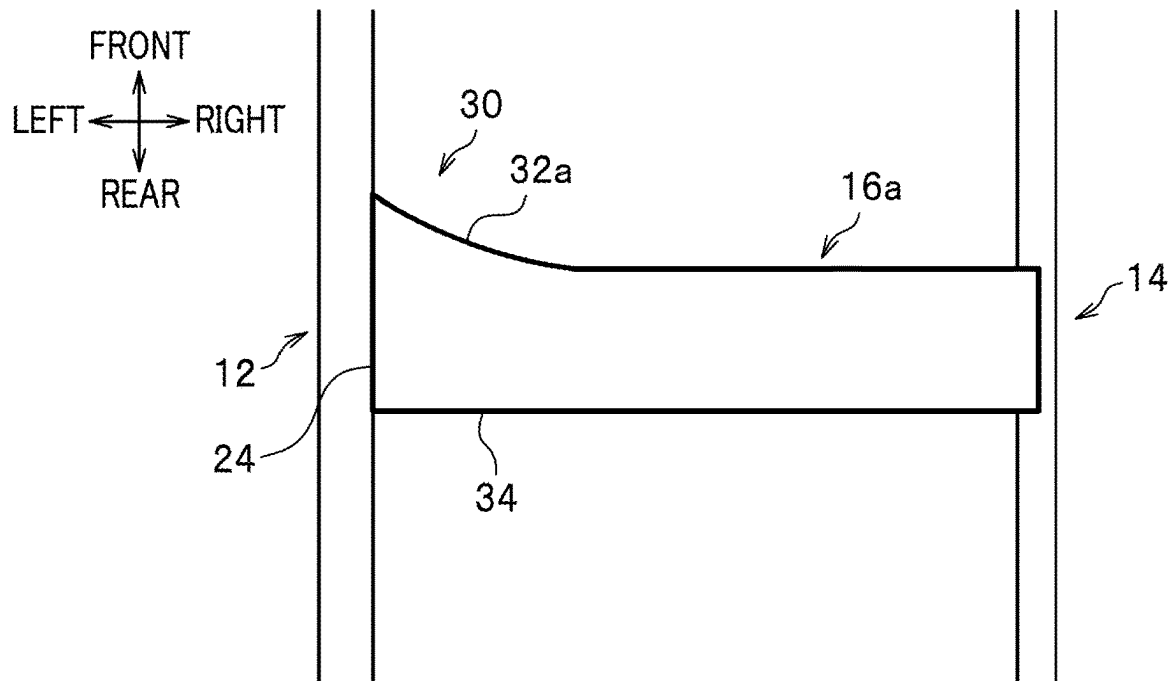


FIG. 8

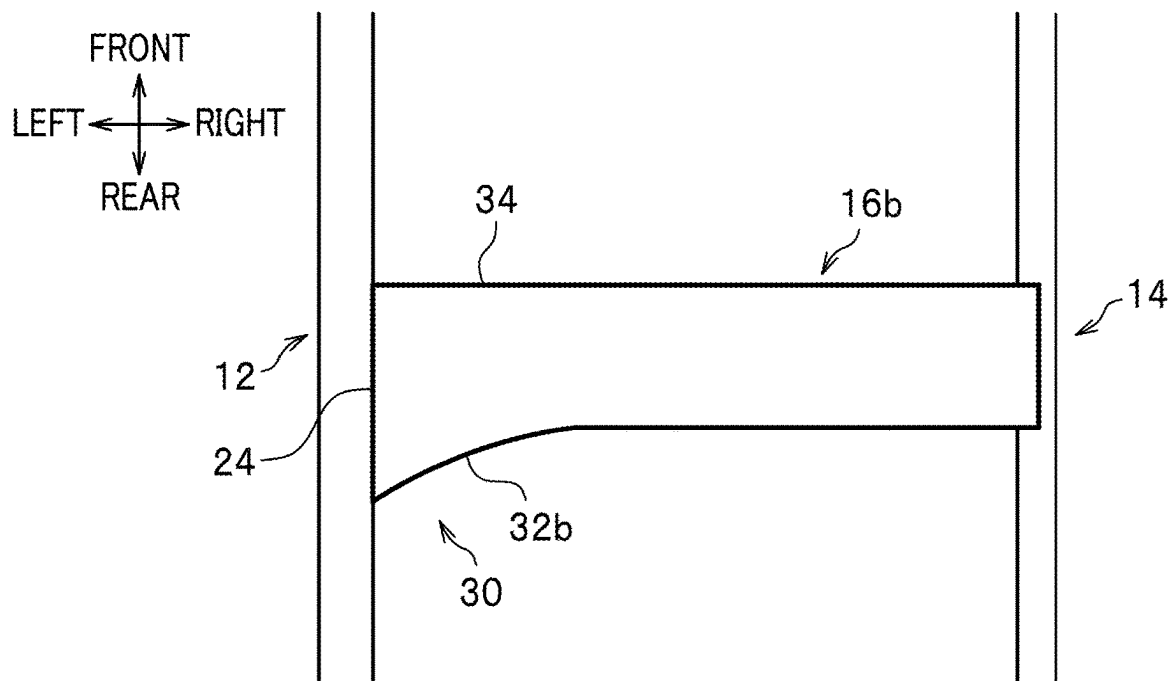


FIG. 9

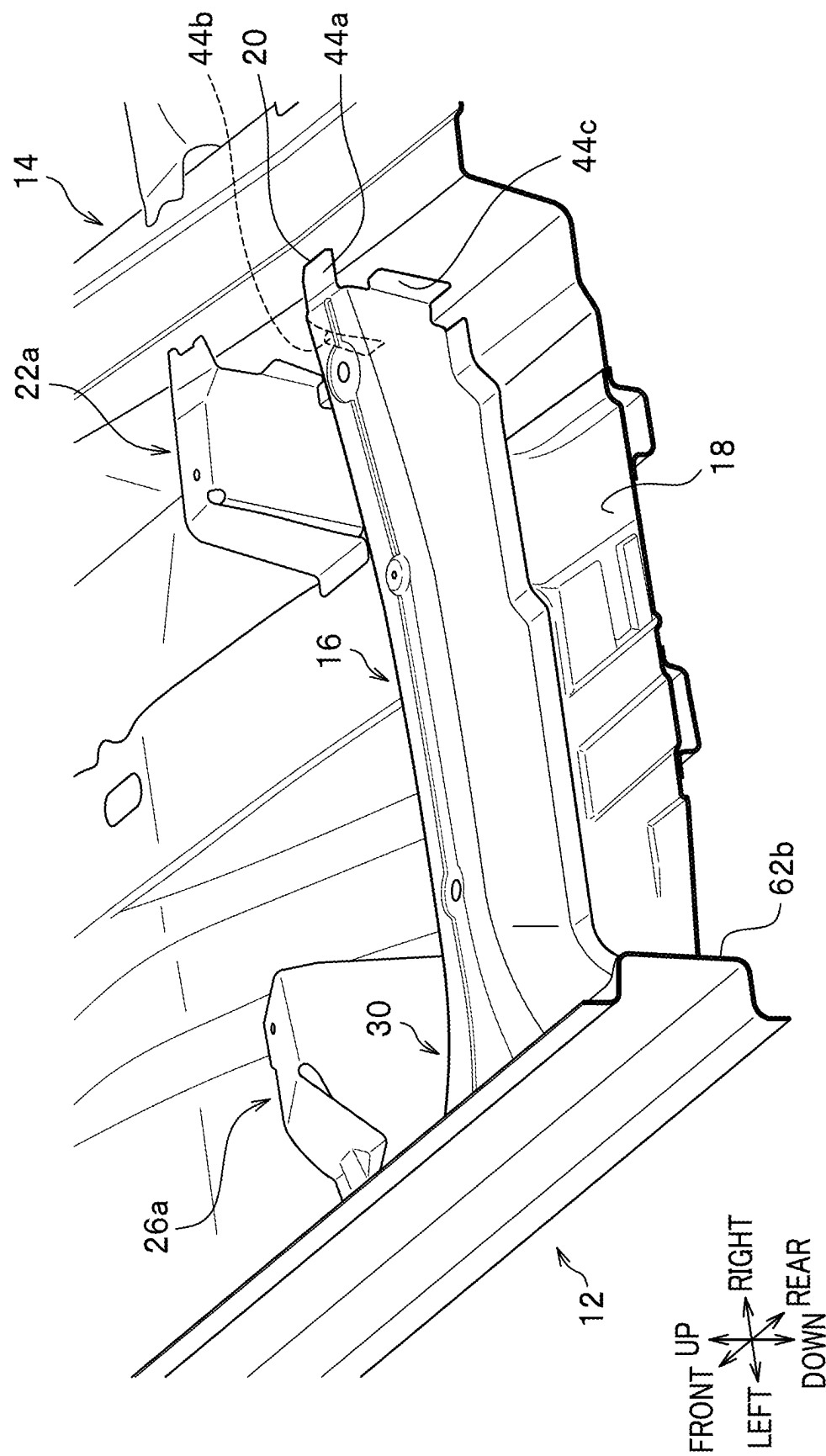


FIG. 10

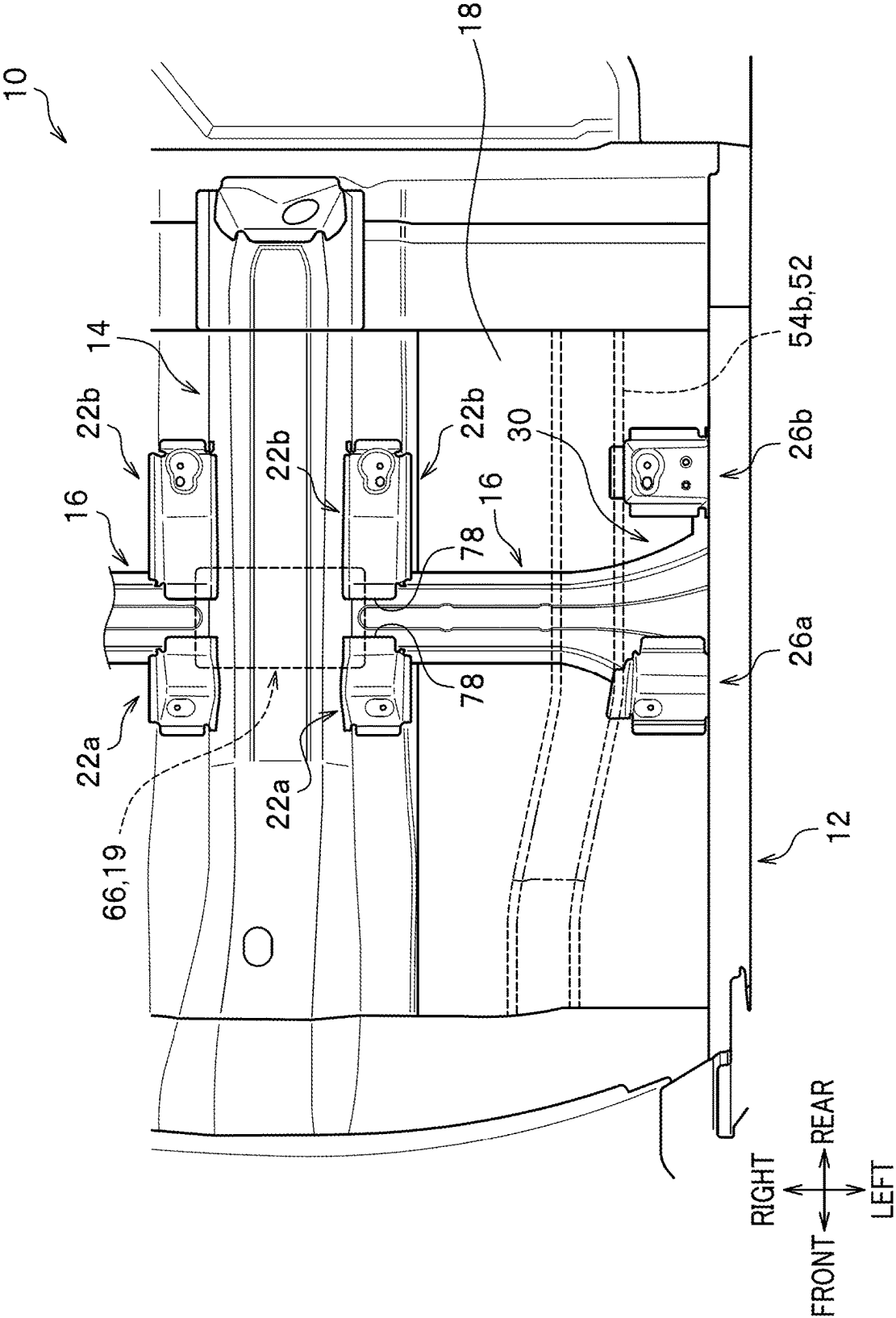


FIG. 11

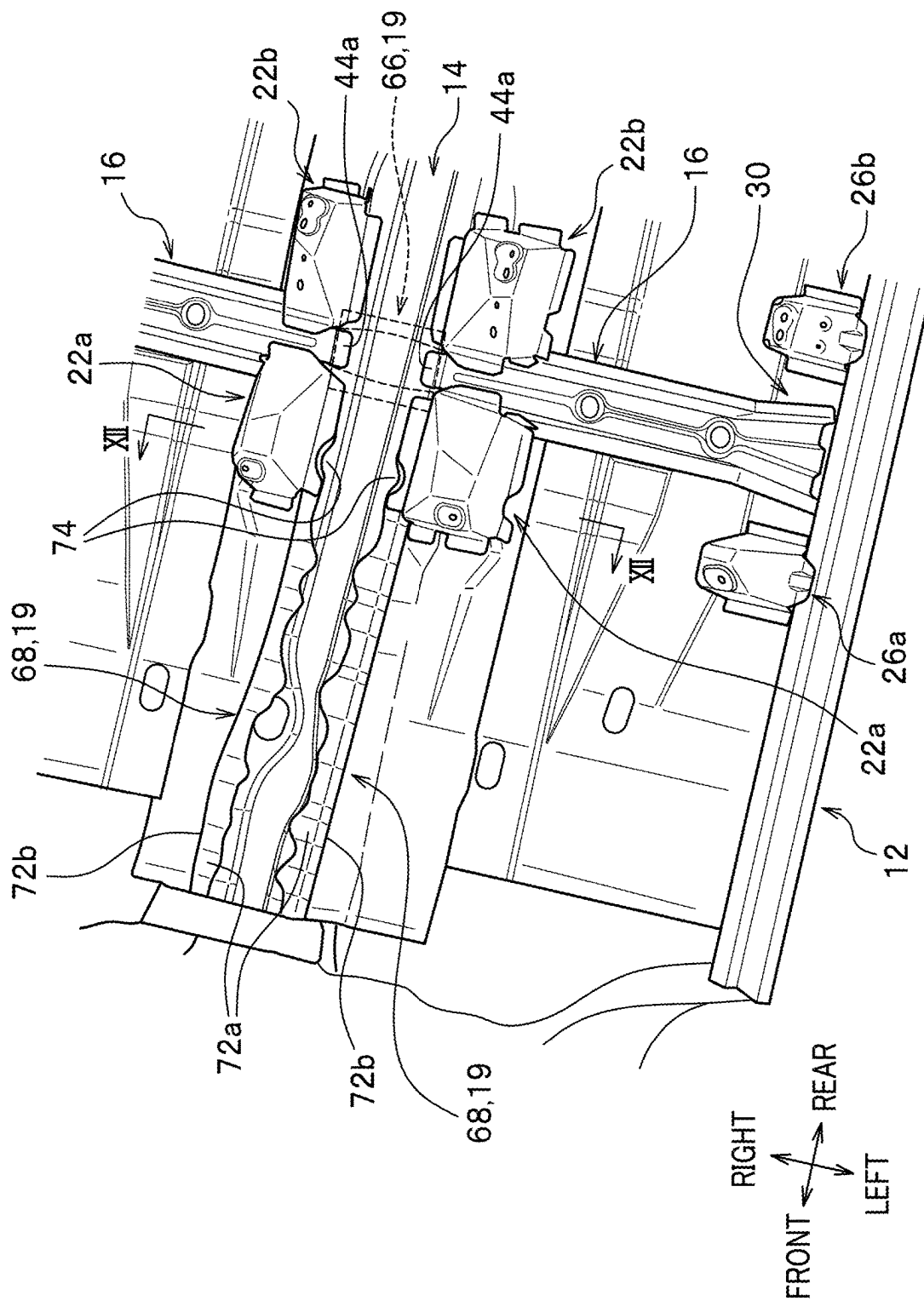


FIG. 12

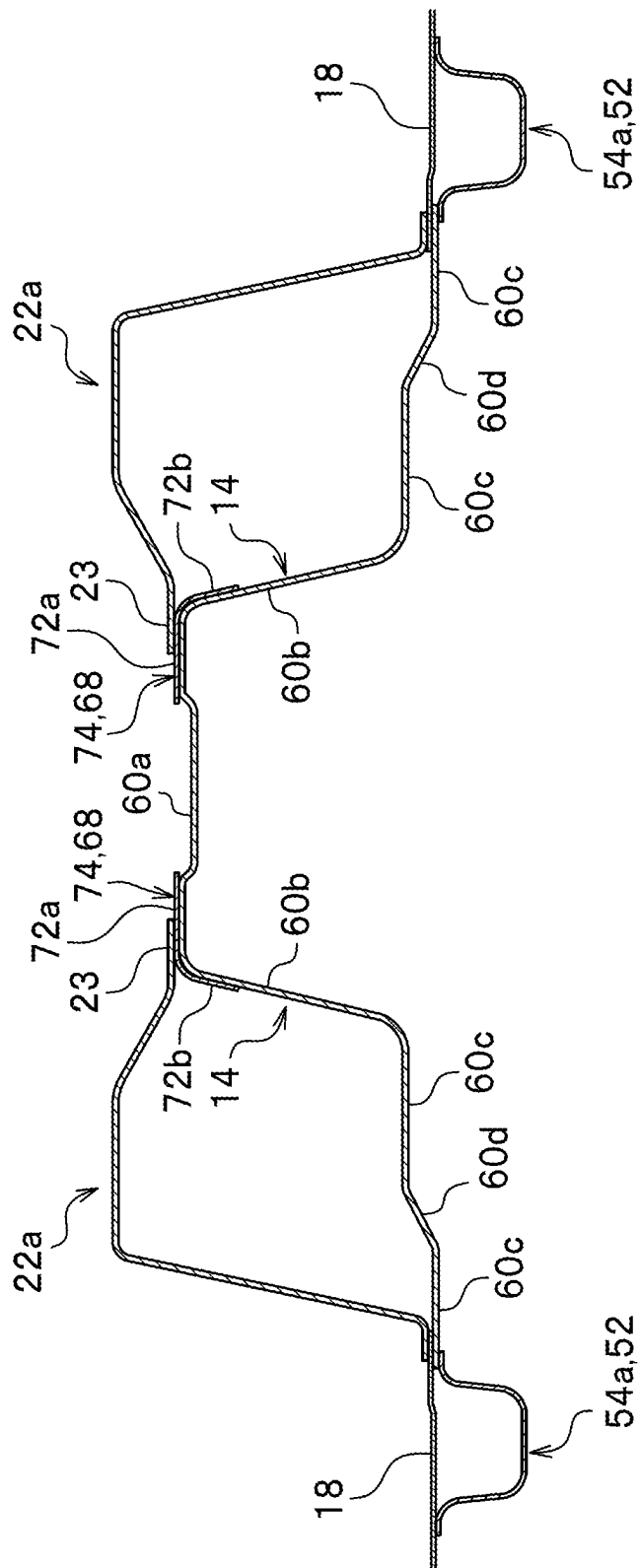


FIG. 13

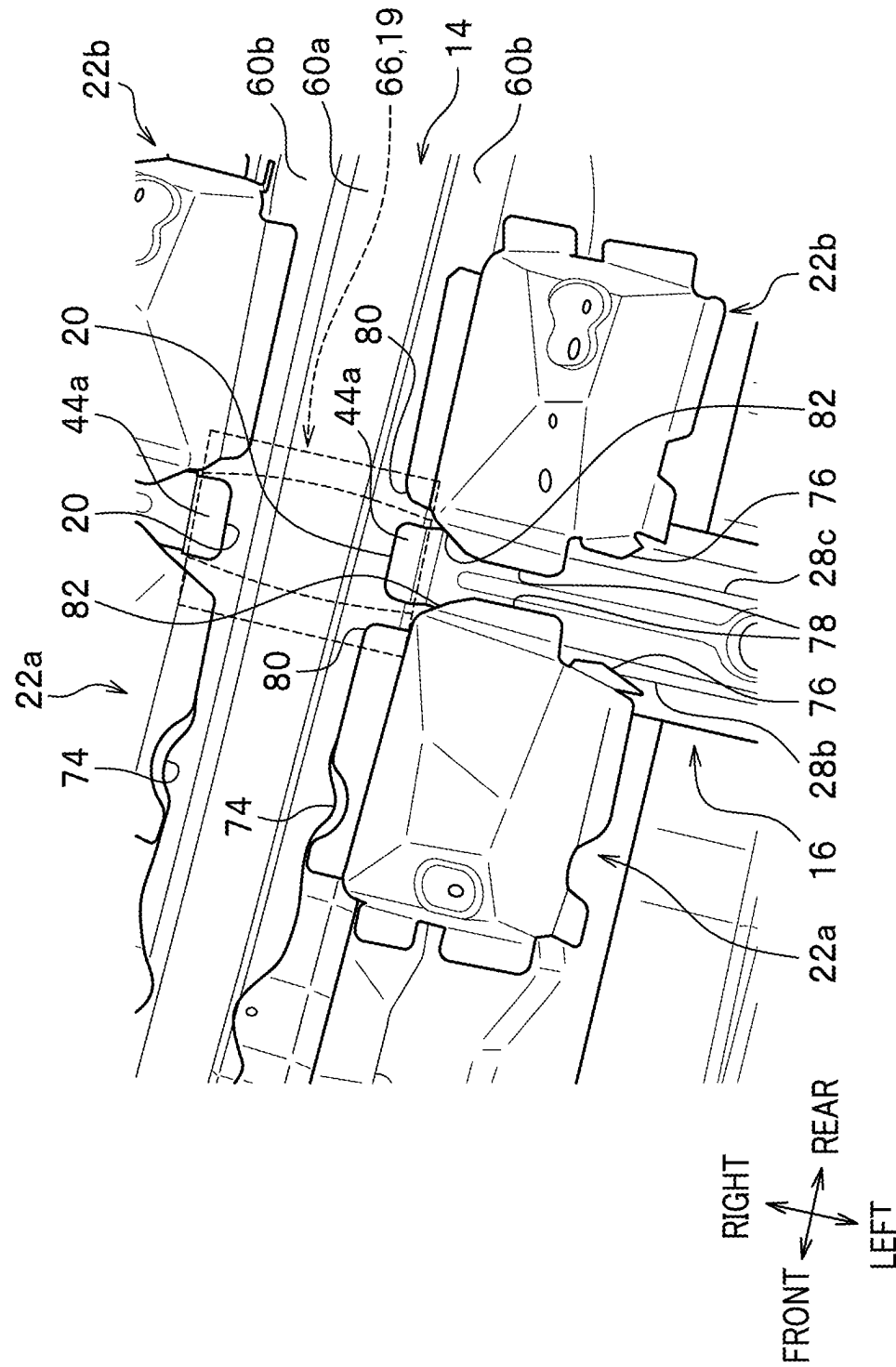
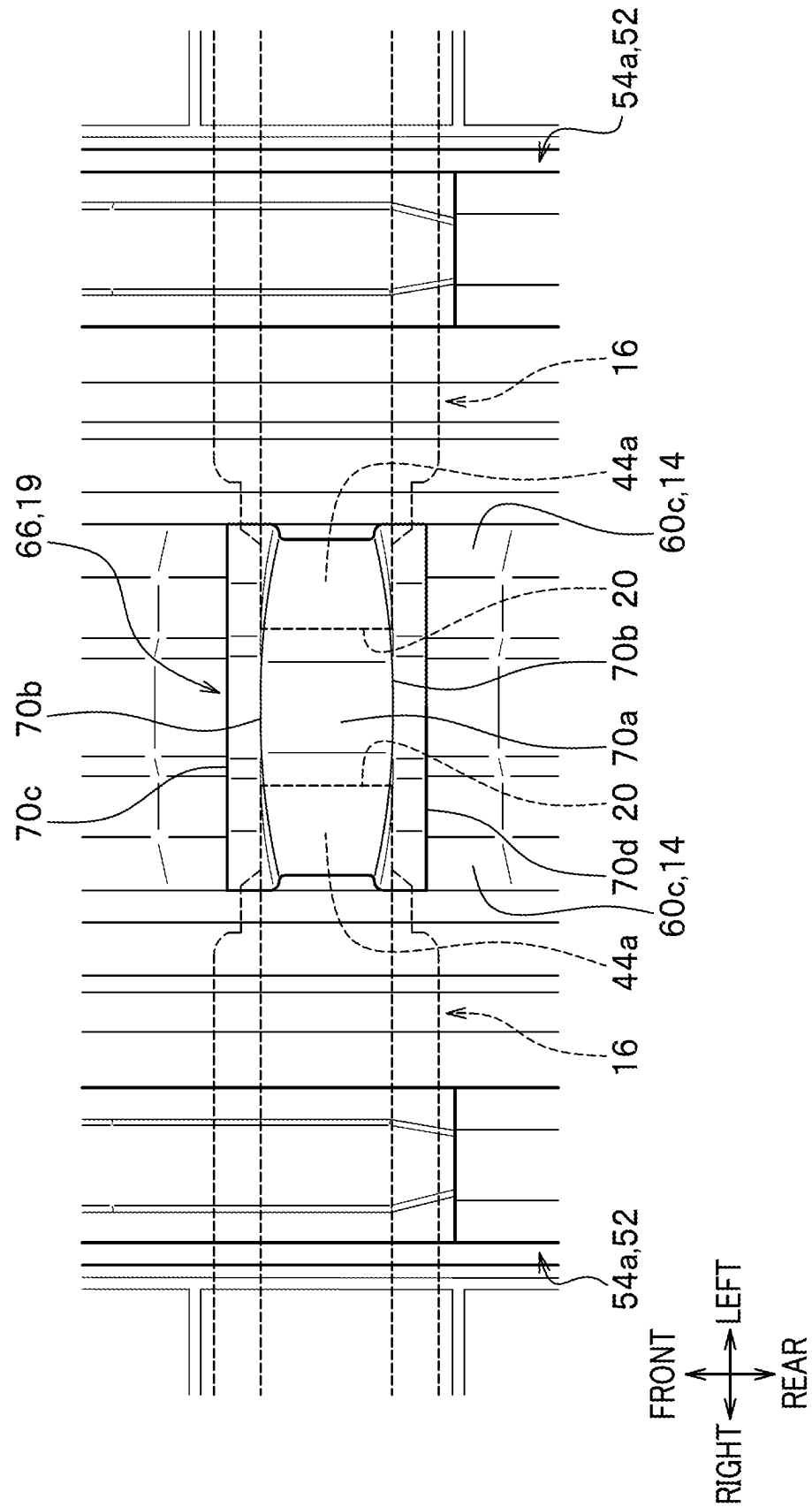


FIG. 14



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LOWER VEHICLE-BODY STRUCTURE

CROSS-REFERENCE TO RELATED APPLICATION

The present application claims the benefit of priority to Japanese Patent Application No. 2021-213756 filed on Dec. 28, 2021, the disclosures of all of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

Field of the Invention

The present invention relates to a lower vehicle-body structure of a vehicle such as an automobile.

Related Art

Japanese Patent No. 6311679 (referred to as Patent Document 1 hereinafter) discloses a lower vehicle-body structure, having an H-shaped structure in a planar view, to include a pair of seat reinforcing members disposed along a side sill and a floor tunnel, and a single floor cross member coupling the pair of seat reinforcing members with each other.

In addition, the lower vehicle-body structure disclosed in Patent Document 1 includes a plurality of tunnel reinforcing members (tunnel cross members) disposed inside the floor tunnel at positions to overlap the seat reinforcing members in a side view along a vehicle width direction. The lower vehicle-body structure disclosed in Patent Document 1 prevents the floor tunnel from being deformed at a time of side collision, with the tunnel reinforcing members.

SUMMARY OF THE INVENTION

However, the lower vehicle-body structure disclosed in Patent Document 1 has a structure in which ends of the floor cross member are abutted on and coupled to substantially the centers of the seat reinforcing members, so that there is a risk that the seat reinforcing members themselves may be deformed when a side collision load is applied, to have seats themselves, attached to the seat reinforcing members, damaged.

The present invention is made in view of the aforementioned problem and is intended to provide a lower vehicle-body structure capable of preventing a floor panel region including a floor tunnel from being deformed.

In addition, the present invention is intended to provide a lower vehicle-body structure capable of improving rigidity of a lower portion of a vehicle body, to enhance safety at a time of collision.

To achieve this objective, the present invention provides a lower vehicle-body structure including: a floor cross member coupling a side sill disposed on an outer side in a vehicle width direction of a vehicle body to a floor tunnel disposed in a center in the vehicle width direction of the vehicle body, and disposed so as to be held between seat brackets in a vehicle front-rear direction; and a tunnel reinforcing member provided in the floor tunnel, wherein the tunnel reinforcing member includes a first tunnel reinforcing member disposed at a position to overlap in the vehicle front-rear direction with the floor cross member, and second tunnel reinforcing members disposed at positions to overlap

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in the vehicle width direction at least partly with the seat bracket, and the seat brackets are coupled to the floor cross member.

The present invention provides a lower vehicle-body structure capable of preventing a floor panel region including a floor tunnel from being deformed, to enhance safety at a time of collision.

In addition, the present invention provides a lower vehicle-body structure capable of improving rigidity of a lower portion of a vehicle body.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a plan view of a front portion of a vehicle applied with a lower vehicle-body structure according to an embodiment of the present invention, as viewed from above;

FIG. 2 is a vertical cross-sectional view, taken along a line II-II in FIG. 1;

FIG. 3 is a partially transparent plan view of the front portion of the vehicle shown in FIG. 1;

FIG. 4 is a partially transparent bottom view of the front portion of the vehicle shown in FIG. 1, as viewed from bottom;

FIG. 5 illustrates a side collision load offset in a vehicle front-rear direction being transmitted;

FIG. 6 is an end view of an inner floor frame and an outer floor frame;

FIG. 7 is a schematic diagram of a modification of a widened portion of a floor cross member;

FIG. 8 is a schematic diagram of another modification of the widened portion of the floor cross member;

FIG. 9 is a partially broken perspective view of an inner end in a vehicle width direction of the floor cross member being coupled to a floor tunnel;

FIG. 10 is a partially transparent plan view of one floor cross member being coupled to the other floor cross member via a first tunnel reinforcing member;

FIG. 11 is a partially enlarged perspective view of a second tunnel reinforcing member;

FIG. 12 is an end view, taken along a line XII-XII in FIG. 11;

FIG. 13 is a partially enlarged view of a first to third flanges provided in a seat bracket; and

FIG. 14 is an enlarged bottom view of the first tunnel reinforcing member shown in FIG. 10.

DESCRIPTION OF THE EMBODIMENTS

Next, a description is given in detail of an embodiment of the present invention, with reference to the drawings as appropriate. In each drawing, “front-rear” refers to a vehicle front-rear direction, “right-left” refers to a vehicle width direction (right-left direction), and “up-down” refers to a vehicle up-down direction (vertically up-down direction).

As shown in FIG. 1, a vehicle 10 applied with a lower vehicle-body structure according to the embodiment of the present invention includes a side sill 12 disposed on an outer side in the vehicle width direction thereof, a floor tunnel 14 disposed in the center of a vehicle body, a single floor cross member 16 disposed along the vehicle width direction and coupling the side sill 12 to the floor tunnel 14, a floor panel 18 disposed between the side sill 12 and the floor tunnel 14, and a tunnel reinforcing member 19 (see FIGS. 11 and 14) provided in the floor tunnel 14 and reinforcing the floor tunnel 14.

An inner end 20 in the vehicle width direction of the floor cross member 16, adjacent to the floor tunnel 14, is disposed

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so as to be held between a pair of first seat brackets **22a** and **22b** in the vehicle front-rear direction. An outer end **24** in the vehicle width direction of the floor cross member **16**, adjacent to the side sill **12**, is disposed so as to be held between a pair of second seat brackets **26a** and **26b** in the vehicle front-rear direction. The first seat brackets **22a** and **22b**, and the second seat brackets **26a** and **26b** are each a member having a box-shape and supporting a seat (not shown) from below.

The floor cross member **16** has a substantially hat-shape in axial cross section (see FIG. 2). As shown in FIG. 2, the floor cross member **16** has an upper wall **28a** to form an upper surface thereof, a front wall **28b** which is bent downward from a front end in the vehicle front-rear direction of the upper wall **28a** and extends along the vehicle width direction, a rear wall **28c** which is bent downward from a rear end in the vehicle front-rear direction of the upper wall **28a** and extends along the vehicle width direction, a front flange **28d** which is bent forward in the vehicle front-rear direction from a lower end of the front wall **28b**, and a rear flange **28e** which is bent rearward in the vehicle front-rear direction from a lower end of the rear wall **28c**. The front flange **28d** and the rear flange **28e** are each joined and fixed to an upper surface of the floor panel **18** by spot welding, for example, or the like.

Further, the outer end **24** in the vehicle width direction of the floor cross member **16** is provided with a widened portion **30** having a wider width in the vehicle front and rear directions toward the side sill **12** (see a portion enclosed by dashed lines in FIG. 1). The widened portion **30** has a front widened portion **32a** which is widened forward in the vehicle front-rear direction and a rear widened portion **32b** which is widened rearward in the vehicle front-rear direction, at the outer end **24** in the vehicle-width direction of the floor cross member **16**. A front end **31** in the vehicle front-rear direction of the widened portion **30** (front widened portion **32a**), which is coupled to the side sill **12**, is disposed forward with respect to a center pillar **48**.

In the present embodiment, the widened portion **30** has both the front widened portion **32a** and the rear widened portion **32b**, but is not limited thereto. For example, as shown in a modification in FIG. 7, the widened portion **30** may only have the front widened portion **32a** which is widened forward in the vehicle front-rear direction at the outer end **24** in the vehicle-width direction of a floor cross member **16a**, and a rear end thereof has a flat portion **34** which is not widened and is continuous to the rest of the floor cross member **16a**. Further, as shown in a floor cross member **16b** of another modification in FIG. 8, the widened portion **30** may only have the rear widened portion **32b** which is widened rearward in the vehicle front-rear direction, and a front end thereof has the flat portion **34** which is not widened and is continuous to the rest of the floor cross member **16b**. Thus, the widened portion **30** includes one having at least one of the front and rear widened portions.

As shown in FIGS. 1 and 2, the upper wall **28a** of the floor cross member **16** is provided, on an upper surface thereof, with a bead **36** which bulges upward and extends along the vehicle width direction. The bead **36** has a front bead **38a** extending along the front wall **28b** and a rear bead **38b** extending along the rear wall **28c**. The front bead **38a** and the rear bead **38b** extend beyond the widened portion **30**, which is positioned on the outer side in vehicle width direction, up to positions (in the vicinity of the floor tunnel **14**) of the first seat brackets **22a** and **22b**, which are disposed inside in the vehicle width direction with respect to the widened portion **30**.

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A recess **40** recessed toward the floor panel **18** (downward) is formed between the front bead **38a** and the rear bead **38b**. The recess **40** extends along a longitudinal direction (vehicle width direction) of the floor cross member **16** and is configured to increase in width in the vehicle front-rear direction at the widened portion **30**. Therefore, the width in the vehicle front-rear direction of the recess **40** closest to the side sill **12** is the largest. The recess **40** is formed, at middle portions thereof, with a plurality of circular recesses **42a** having a large, middle, and small inner diameters, respectively and a single oval recess **42b**, for example. Note that the width of the recess **40**, excluding the recess **40** formed at the widened portion **30**, the circular recesses **42a**, and the oval recess **42b**, is formed substantially constant. Note that, in the present embodiment, the recess **40** is formed between the front bead **38a** and the rear bead **38b** as an example, but is not limited thereto, and a convex portion which bulges upward may be formed in place of the recess **40**.

The inner end **20** in the vehicle width direction of the floor cross member **16** is coupled to the floor tunnel **14**. That is, as shown in FIG. 9, the inner end **20** in the vehicle width direction of the floor cross member **16** has an upper flange **44a**, which is joined to an upper surface of the floor tunnel **14**, and a front flange **44b** and a rear flange **44c**, which are joined to a side surface of the floor tunnel **14**.

In addition, the first seat brackets **22a** and **22b**, coupled to the floor tunnel **14** are provided at an end of the floor cross member **16** (a portion adjacent to the inner end **20** in the vehicle width direction), closer to the floor tunnel **14**. The recess **40** formed in the upper surface of the floor cross member **16** extends up to a position to overlap in the vehicle width direction with the first seat brackets **22a** and **22b** (see FIG. 1).

Further, the outer end **24** in the vehicle width direction of the floor cross member **16** has a front flange **46a** and a rear flange **46b** which are joined to the side sill **12** (inner side sill **62b**) (see FIG. 2). In the vehicle front-rear direction, the rear flange **46b** is joined (coupled) to the side sill **12** at a position to overlap a front end **50** in the vehicle front-rear direction of the center pillar **48**, and the front flange **46a** is joined (coupled) to the side sill **12** at a forward position in the vehicle front-rear direction with respect to the center pillar **48**.

The floor panel **18** is provided, at a lower surface thereof to face the ground, with a floor frame **52** extending along the vehicle front-rear direction. As shown in FIG. 4, the floor frame **52** has an inner floor frame **54a** closer to the floor tunnel **14** and an outer floor frame **54b** closer to the side sill **12**. The inner floor frame **54a** is disposed linearly along the floor tunnel **14**. The outer floor frame **54b** has a rear portion with respect to the floor cross member **16**, which is disposed linearly along the side sill **12**, and a front portion with respect to the floor cross member **16**, which is bent inward in the vehicle width direction.

As shown in FIGS. 3 and 4, the floor cross member **16** is disposed to intersect the floor frame **52** (inner floor frame **54a** and outer floor frame **54b**). In addition, the widened portion **30** of the floor cross member **16** is widened from a position to overlap in the vehicle up-down direction with (intersect) the outer floor frame **54b** toward an outer side in the vehicle width direction (toward the side sill **12**), in planar view (see FIG. 3).

Further, at least a portion (such as the rear wall **28c** or the like) of the floor cross member **16** is disposed at a position to overlap in the vehicle front-rear direction (as viewed in

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the vehicle width direction) with the front end **50** in the vehicle front-rear direction provided at a lower end of the center pillar **48** (see FIG. 2).

As shown in FIG. 6, the floor frame **52** has a hat-shape in axial cross section, and has a lower wall **56a**, a pair of side walls **56b** continuing to the lower wall **56a** and facing each other, and flanges **56c** continuing to upper ends of the side walls **56b**. The flanges (front flange **28d** and rear flange **28e**) of the floor cross member **16** and the flanges **56c** of the floor frame **52** are welded and fixed to the upper surface and lower surface of the floor panel **18**, respectively.

In other words, the floor panel **18** has the flanges (front flange **28d** and rear flange **28e**) of the floor cross member **16** joined to the upper surface thereof (see FIG. 2), and the flanges **56c** of the floor frame **52** joined to the lower surface thereof (see FIG. 6). In this case, three members are overlapped in the up-down direction with each other and welded simultaneously and integrally at the intersection **58** at which the flanges (front flange **28d** and rear flange **28e**) of the floor cross member **16** vertically intersects the flanges **56c** of the floor frame **52** (see joint points at the intersection **58** in FIG. 3).

Further, as shown in FIG. 6, the floor panel **18** has flanges **72a** of the first seat brackets **22a** and **22b** (second seat brackets **26a** and **26b**) joined to the upper surface thereof, and flanges **56c** of the floor frame **52** (**54a** and **54b**) joined to the lower surface thereof. In this case, the three members of the seat bracket (**22a**, **22b**, **26a**, and **26b**), the floor panel **18**, and the floor frame **52** are overlapped in the up-down direction with each other and spot welded simultaneously and integrally. Note that glue G is placed in a welded region between the flange **72a** and the upper surface of the floor panel **18** (see FIG. 6).

As shown in FIG. 12, the floor tunnel **14** has a hat-shape in axial cross section. The floor tunnel **14** has a pair of tunnel side walls **60b** facing the side sill **12**, a tunnel upper wall **60a** connecting upper ends of the pair of tunnel side walls **60b**, and a pair of tunnel lower walls **60c** bent from lower ends of the tunnel side walls **60b** and extending along the vehicle width direction. The tunnel lower wall **60c** is provided in the middle thereof with an inclined wall **60d** inclining downward from an inner side in the vehicle width direction thereof toward an outer side in the vehicle width direction thereof.

The tunnel reinforcing member **19** has a first tunnel reinforcing member **66** (see FIG. 14) provided outside a vehicle compartment so as to face the ground and second tunnel reinforcing members **68** (see FIG. 11) provided inside the vehicle compartment. In the present embodiment, the first tunnel reinforcing member **66** is provided outside the vehicle compartment and the second tunnel reinforcing members **68** are provided inside the vehicle compartment, but the present invention is not limited thereto and said members may be provided either inside or outside the vehicle compartment.

As shown in FIG. 14, the first tunnel reinforcing member **66** has a rectangular shape in bottom view and has a hat-shape in cross section along the vehicle front-rear direction. The first tunnel reinforcing member **66** is disposed between one floor cross member **16** (on the left side in a planar view, for example) and the other floor cross member **16** (on the right side in a planar view, for example), which face each other along the vehicle width direction, to couple the one floor cross member **16** with the other floor cross member **16** in the middle in the vehicle width direction (see FIG. 10), as described below. In other words, the first tunnel reinforcing member **66** is disposed at a position to overlap

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in the vehicle front-rear direction with the right and left floor cross members **16**. In still other words, the first tunnel reinforcing member **66** is aligned in the vehicle width direction with the right and left floor cross members **16**.

As shown in FIG. 14, the first tunnel reinforcing member **66** has a bottom wall **70a** facing the ground, a pair of side walls **70b** bent upward from a front end and a rear end in the vehicle front-rear direction of the bottom wall **70a** and facing each other, and a front joining flange **70c** and a rear joining flange **70d**, which respectively extend forward and rearward in the vehicle front-rear direction from a front end and a rear end of the pair of side walls **70b**.

The front joining flange **70c** and the rear joining flange **70d** of the first tunnel reinforcing member **66** are respectively joined to a lower surface of the tunnel lower wall **60c** of the floor tunnel **14** by spot welding, for example, or the like. Further, the inner end **20** in the vehicle width direction of the floor cross member **16** is coupled to the floor tunnel **14**. That is, the upper flange **44a** of the floor cross member **16** is joined to the tunnel upper wall **60a** of the floor tunnel **14**, and the front flange **44b** and the rear flange **44c** of the floor cross member **16** are joined to the tunnel side walls **60b** of the floor tunnel **14**, respectively. As a result, the first tunnel reinforcing member **66** is coupled to the floor cross member **16** via the floor tunnel **14**.

As shown in FIGS. 11 and 12, each of the second tunnel reinforcing members **68** integrally includes an upper wall **72a** having a substantially hat-shape in axial cross section and joined to the tunnel upper wall **60a** of the floor tunnel **14**, and a side wall **72b** bent downward from an outer end of the upper wall **72a** and joined to the tunnel side wall **60b** of the floor tunnel **14**. The second tunnel reinforcing members **68** extend in the vehicle front-rear direction along a ridge of the floor tunnel **14** and are disposed to face each other along the vehicle width direction.

In addition, a rear portion **74** in the vehicle front-rear direction, which is at least a portion of each of the second tunnel reinforcing members **68**, is disposed at a position to overlap in the vehicle width direction with the first seat bracket **22a** on the front side (see FIG. 12). That is, the rear portion **74** in the vehicle front-rear direction of the second tunnel reinforcing member **68** is interposed between an inner end **23** in the vehicle width direction of the first seat bracket **22a** and the tunnel upper wall **60a** of the floor tunnel **14**, and three members of the rear portion **74** in the vehicle front-rear direction of the second tunnel reinforcing member **68**, the inner end **23** in the vehicle width direction of the first seat bracket **22a**, and the tunnel upper wall **60a** of the floor tunnel **14**, are joined integrally. Accordingly, the first seat bracket **22a** on the front side is joined (coupled) to the second tunnel reinforcing members **68** which extend in the vehicle front-rear direction across the tunnel side walls **60b** and the tunnel upper wall **60a**.

As shown in FIG. 13, the first seat brackets **22a** and **22b** disposed closer to the floor tunnel **14** each have a first flange **76**, a second flange **78**, and a third flange **80**. The first flange **76** is positioned on an outer side in the vehicle width direction and joined to both the front wall **28b** and the rear wall **28c** of the floor cross member **16**. The second flange **78** is positioned between the first flange **76** and the third flange **80** and joined to the upper wall **28b** of the floor cross member **16**. The third flange **80** is positioned on an inner side in the vehicle width direction with respect to the second flange **78** and joined to the first tunnel reinforcing member **66** via the floor tunnel **14**. The first seat brackets **22a** and **22b** are coupled (joined) to the floor cross member **16** via the first

flanges 76, the second flanges 78, and portions (rear end of the former and front end of the latter, respectively) of the third flanges 80.

The second flange 78 positioned on a front side of the first seat bracket 22b and the second flange 78 positioned on a rear side of the seat bracket 22a are disposed apart from each other in the vehicle front-rear direction (see FIG. 10). In addition, the front and rear second flanges 78 are each provided, at an inner portion in the vehicle width direction thereof, with an inclined portion 82 inclining more in the vehicle front-rear direction toward the floor tunnel 14.

The second seat brackets 26a and 26b, disposed closer to the side sill 12, are joined to the upper surface of the floor panel 18 by spot welding, for example, or the like. The floor panel 18 has the floor frame 52 joined thereto, on the lower surface thereof, so as to intersect the floor cross member 16. Accordingly, the second seat brackets 26a and 26b are joined to the floor frame 52 via the floor panel 18.

The side sill 12 has an outer side sill 62a (see FIG. 1) disposed on the outer side in the vehicle width direction and an inner side sill 62b (see FIG. 9) disposed on the inner side in the vehicle width direction. In a cross section in the axial direction of the side sill 12, a reinforcing member which cross-sectionally reinforces the side sill 12 is provided between the outer side sill 62a and the inner side sill 62b.

The vehicle 10 applied with the lower vehicle-body structure of the present embodiment is basically formed as described above, and advantageous effects thereof are described below.

The present embodiment includes: the floor cross member 16 coupling the side sill 12 disposed on the outer side in the vehicle width direction of the vehicle body to the floor tunnel 14 disposed in the center in the vehicle width direction of the vehicle body, and disposed so as to be held between the first seat brackets 22a and 22b in the vehicle front-rear direction; and the tunnel reinforcing member 19 provided in the floor tunnel 14. The tunnel reinforcing member 19 includes the first tunnel reinforcing member 66 disposed at a position to overlap in the vehicle front-rear direction with the floor cross member 16, and the second tunnel reinforcing members 68 disposed at positions to overlap in the vehicle width direction with the first seat brackets 22a and 22b. The first seat brackets 22a and 22b are joined (coupled) to the floor cross member 16.

The present embodiment includes the first seat brackets 22a and 22b to be joined to the floor cross member 16 so as to be aligned in the vehicle front-rear direction and the first tunnel reinforcing member 66 to overlap in the vehicle front-rear direction with the floor cross member 16, so that a side impact load applied at the time of side collision can be smoothly transmitted from the floor cross member 16 to the first tunnel reinforcing member 66. In addition, the side impact load can be transmitted and dispersed to the second tunnel reinforcing members 68 via the first seat brackets 22a and 22b. Accordingly, in the present embodiment, the floor tunnel 14 and the first seat brackets 22a and 22b, even when the side impact load is transmitted thereto via the floor cross member 16, are prevented from being deformed. As a result, the present embodiment can prevent a floor panel region including the floor tunnel 14 from being deformed. In addition, the present embodiment can improve rigidity of a lower portion of the vehicle body.

Further, in the present embodiment, the first tunnel reinforcing member 66 is coupled to the floor cross member 16 via the floor tunnel 14.

In the present embodiment, the first tunnel reinforcing member 66 being joined to the floor cross member 16 allows

for transmitting the side impact load applied to one floor cross member 16 to the other floor cross member 16 via the first tunnel reinforcing member 66. Accordingly, floor tunnel 14 is prevented from being deformed due to the side impact load. In addition, in the present embodiment, the side impact load can be dispersed by the first seat brackets 22a and 22b, and the floor cross member 16, so that the first seat brackets 22a and 22b, and the floor cross member 16 are prevented from being deformed.

Further, in the present embodiment, the floor tunnel 14 has the pair of tunnel side walls 60b facing the side sill 12 and the tunnel upper wall 60a connecting the upper ends of the tunnel side walls 60b with each other. The first seat bracket 22a on the front side is joined (coupled) to the second tunnel reinforcing members 68 which extend in the vehicle front-rear direction across the tunnel side walls 60b and the tunnel upper wall 60a.

In the present embodiment, the first seat bracket 22a on the front side is joined to the second tunnel reinforcing members 68, so that the side impact load transmitted from the floor cross member 16 to the first seat bracket 22a on the front side can be dispersed over the entire floor tunnel 14. Accordingly, the floor panel region (floor panel 18 and floor tunnel 14) is prevented from being deformed due to the side impact load.

Moreover, in the present embodiment, the floor cross member 16 has the upper wall 28a to form the upper surface thereof, the front wall 28 on the front side in the vehicle front-rear direction, and the rear wall 28c on the rear side in the vehicle front-rear direction. The first seat brackets 22a and 22b have the first flanges 76 joined to the front wall 28b and the rear wall 28c of the floor cross member 16, respectively, the second flanges 78 joined to the upper wall 28a, and the third flanges 80 positioned on the inner side in the vehicle width direction with respect to the second flanges 78. The third flanges 80 have the front end and rear end thereof, as at least portions thereof, coupled to the first tunnel reinforcing member 66 via the floor tunnel 14.

In the present embodiment, the first seat brackets 22a and 22b are joined to the floor cross member 16 via the first flanges 76 and the second flanges 78, and the third flanges 80 are coupled to the first tunnel reinforcing member 66, so that a side impact load can be transmitted to the first tunnel reinforcing member 66 while the side impact load applied to the floor cross member 16 is dispersed to the first seat brackets 22a and 22b, and the second tunnel reinforcing members 68. As a result, the present embodiment can further prevent the floor panel region including the floor tunnel 14 from being deformed.

Further, in the present embodiment, the second flanges 78 positioned on the front side and the rear side in the vehicle front-rear direction of the first seat brackets 22a and 22b are disposed apart from each other in the vehicle front-rear direction and have the inclined portions 82 which are more inclined in the vehicle front-rear direction toward the floor tunnel 14.

In the present embodiment, the second flanges 78 are disposed apart from each other and have the inclined portions 82 which incline in the vehicle front-rear direction, so that a side impact load applied from the floor cross member 16 can be transmitted along the vehicle front-rear direction (longitudinal direction of the floor tunnel 14). Thus, in the present embodiment, the side impact load applied to the first tunnel reinforcing member 66 is dispersed in the vehicle front-rear direction, to prevent the floor panel region including the floor tunnel 14 from being deformed.

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Further, in the present embodiment, the seat bracket include the first seat brackets **22a** and **22b** disposed closer to the floor tunnel **14**, and the second seat brackets **26a** and **26b** disposed closer to the side sill **12**. The floor frame **52** disposed at a position intersecting the floor crossmember **16** is provided on the lower surface of the floor panel **18**. The second seat brackets **26a** and **26b** are coupled to the floor frame **52** via the floor panel **18**.

In the present embodiment, the second seat brackets **26a** and **26b** closer to the side sill **12** are joined to the floor frame **52** intersecting the floor cross member **16**, so that an offset side impact load applied to one floor cross member **16** can efficiently be transmitted to the other floor cross member **16** via the floor frame **52**. As a result, in the present embodiment, the offset side impact load is dispersed inside in the vehicle width direction of the floor crossmember **16**, to prevent the floor panel region including the floor tunnel **14** from being deformed.

Further, in the present embodiment, the floor cross member **16** is disposed at a position to at least partly overlap in the vehicle front-rear direction with the front end **50** in the vehicle front-rear direction provided at the lower end of the center pillar **48**.

In the present embodiment, the floor cross member **16** is disposed at a position to overlap in the vehicle front-rear direction with the front end **50** in the vehicle front-rear direction of the center pillar **48**, so that a side impact load applied from the center pillar **48** and an offset side impact load applied from a forward position with respect to the center pillar **48** can be efficiently transmitted to the floor cross member **16**, respectively. As a result, in the present embodiment, the side impact load (including the offset side impact load) transmitted to the floor cross member **16** can be dispersed to the floor tunnel **14**, and the first seat brackets **22a** and **22b**, disposed inside in the vehicle width direction of the floor cross member **16**.

Further, in the present embodiment, the outer end **24** in the vehicle width direction of the floor cross member **16** is provided with the widened portion **30** having a wider width in the vehicle front and rear directions toward the side sill **12**. The front end **31** in the vehicle front-rear direction of the widened portion **30**, which is coupled to the side sill **12**, is disposed forward with respect to the center pillar **48**.

In the present embodiment, the widened portion **30**, which is formed by the outer end **24** in the vehicle width direction of the floor cross member **16** widened in the vehicle front and rear directions, is provided, so that, even when a side impact load is applied to a position offset in the vehicle front-rear direction with respect to the floor cross member **16**, an offset side impact load can be efficiently transmitted along the widened portion **30** of the floor cross member **16** (see an open arrow A in FIG. 5). As a result, the present embodiment can prevent the floor panel region including the floor tunnel **14** from being deformed due to the side impact load offset in the vehicle front-rear direction.

Further, in the present embodiment, load transmission efficiency is improved for a side impact load (including an offset side impact load), so that arranging multiple floor cross members **16** on one side is not necessary, and it is sufficient to provide a single floor cross member **16** on the one side. Accordingly, the present embodiment achieves reducing the lower vehicle-body in weight, improves energy efficiency, and increases a leg room for occupants.

What is claimed is:

1. A lower vehicle-body structure, comprising:
 - a floor cross member coupling a side sill disposed on an outer side in a vehicle width direction of a vehicle body

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to a floor tunnel disposed in a center in the vehicle width direction of the vehicle body, and disposed so as to be held between seat brackets in a vehicle front-rear direction; and

a tunnel reinforcing member provided in the floor tunnel, wherein

the tunnel reinforcing member includes a first tunnel reinforcing member disposed at a position to overlap in the vehicle front-rear direction with the floor cross member, and second tunnel reinforcing members disposed at positions to overlap in the vehicle width direction at least partly with the seat bracket,

the seat brackets include seat brackets adjacent to the floor tunnel and seat brackets adjacent to the side sill, and the seat brackets adjacent to the floor tunnel are coupled to the floor cross member,

the first tunnel reinforcing member is coupled to the floor cross member via the floor tunnel,

the floor tunnel has tunnel side walls facing the side sill, and a tunnel upper wall connecting upper ends of the tunnel side walls,

the seat brackets adjacent to the floor tunnel include front seat brackets disposed forward of the floor cross member in the vehicle front-rear direction, and rear seat brackets disposed rearward of the floor cross member in the vehicle front-rear direction, each of the front seat brackets and the rear seat brackets including a seat bracket disposed on each side of the tunnel in the vehicle width direction, and the front seat brackets are coupled to the second tunnel reinforcing members, which extend in the vehicle front-rear direction across the tunnel side walls and the tunnel upper wall,

the floor cross member has an upper wall to form an upper surface thereof, a front wall on a front side thereof in the vehicle front-rear direction, and a rear wall on a rear side thereof in the vehicle front-rear direction,

the front seat brackets each have a first flange coupled to the front wall and the rear seat brackets each have a first flange coupled to the rear wall of the floor cross member, and each of the front seat brackets and the rear seat brackets have a second flange coupled to the upper wall, and a third flange positioned on an inner side in the vehicle width direction with respect to the second flange, and

the third flange has at least a portion thereof coupled to the first tunnel reinforcing member via the floor tunnel.

2. The lower vehicle-body structure according to claim 1, wherein the second flanges respectively positioned on a front side and a rear side in the vehicle front-rear direction of the seat brackets are disposed apart from each other in the vehicle front-rear direction and have inclined portions which are more inclined in the vehicle front-rear direction toward the floor tunnel.

3. A lower vehicle-body structure, comprising:

a floor cross member coupling a side sill disposed on an outer side in a vehicle width direction of a vehicle body to a floor tunnel disposed in a center in the vehicle width direction of the vehicle body, and disposed so as to be held between seat brackets in a vehicle front-rear direction; and

a tunnel reinforcing member provided in the floor tunnel, wherein

the tunnel reinforcing member includes a first tunnel reinforcing member disposed at a position to overlap in the vehicle front-rear direction with the floor cross member, and second tunnel reinforcing members dis-

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posed at positions to overlap in the vehicle width direction at least partly with the seat brackets, the seat brackets are coupled to the floor cross member, the first tunnel reinforcing member is coupled to the floor cross member via the floor tunnel, and

the floor cross member is disposed at a position to at least partly overlap in the vehicle front-rear direction with a front end in the vehicle front-rear direction of a center pillar.

4. The lower vehicle-body structure according to claim 3, wherein

an outer end in the vehicle width direction of the floor cross member is provided with a widened portion having a wider width in the vehicle front-rear direction toward an outer side in the vehicle width direction, and a front end in the vehicle front-rear direction of the widened portion, which is coupled to the side sill, is disposed forward with respect to the center pillar.

5. The lower vehicle-body structure according to claim 3, wherein

the floor tunnel has tunnel side walls facing the side sill, and a tunnel upper wall connecting upper ends of the tunnel side walls, and

the seat brackets include front seat brackets disposed forward of the floor cross member in the vehicle front-rear direction, the front seat brackets including a

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seat bracket disposed on each side of the tunnel in the vehicle width direction, and the front seat brackets are coupled to the second tunnel reinforcing members, which extend in the vehicle front-rear direction across the tunnel side walls and the tunnel upper wall.

6. The lower vehicle-body structure according to claim 5, wherein

the floor cross member has an upper wall to form an upper surface thereof, a front wall on a front side thereof in the vehicle front-rear direction, and a rear wall on a rear side thereof in the vehicle front-rear direction,

the seat brackets each have a first flange coupled to one of the front wall and the rear wall of the floor cross member, a second flange coupled to the upper wall, and a third flange positioned on an inner side in the vehicle width direction with respect to the second flange, and the third flange has at least a portion thereof coupled to the first tunnel reinforcing member via the floor tunnel.

7. The lower vehicle-body structure according to claim 6, wherein the second flanges respectively positioned on a front side and a rear side in the vehicle front-rear direction of the seat brackets are disposed apart from each other in the vehicle front-rear direction and have inclined portions which are more inclined in the vehicle front-rear direction toward the floor tunnel.

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